

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MECH 511 — Past Questions & Materials
by Renance

Course: MECH 511
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — MECH 511

Level 300 • Second Semester • PastQuestion

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COLLEGE OF TECHNOLOGY @ ENGINEERING SOURCES

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2024/2025 Session

urse & Title: MEE 312 Mechanics of Machines Time Allowed: 2 Hours

truction: Answer four (4) questions.

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ESTION ONE

What distinguishes a machine from a structure?

Determine the degrees of freedom for the specified mechanism in Figure 1.

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igure 1

'ESTION TWO {

Illustrate four-bar chain inversions with sketches.

State Grashof's law and its limitat'ons.

ESTION THREE

Define: (i) Degree of freedom, (ii) Kinematic chain} (iii) Higher pair, (iv)Quaternary joint.

List and explain ten (10) gear terninologies or "oe with a good sketch.

ESTION FOUR |

What is a Cam? Identify five applications of Cams

For the mechanism shown in the "igure, calculate its degrees of freedom (F).

ESTION FIVE |

: gearing of a machine tool is shown in Fig. 1. The motor shaft is connected to gear A and rotates 75 r.p.m. The gear wheels B, C, D and E are fixed {o parallel shafts rotaling together. The final gear F is

d on the output shaft. What is the speed of gear F? The number of teeth on each gear is given below.

Gear | va | B |c.

—_ = a — — D

Number of , | = | : |

Teeth | - a 20 | 50 | 2s 15 26 | 65 |

— — — — — — — — 1 — |

Aa- ----~ale, '\

(a+ |) (E+ () 4

B

D -

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_Gear _ A B | ; |

Number of f i | —_ : =

T

eelh 20 joe | 25 | 75 | 26 | ~ =65

* y (a ~

. {ff fw~ ey, \ \

(Co |) (e+ () +)

D -F

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Figure 1

QUESTION TWO

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State Grashof's law and its limitations.

QUESTION THREE

Define: (i) Degree of freedom, (ii) Kinematic chain, (iii) Higher pair, (iv) Quaternary joint.

List and explain ten (10) gear terminologies or nomenclatures with a good sketch.

What is a Cam? Identify five applications of Cams

For the mechanism shown in the figure, calculate its degrees of freedom (F).

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QUESTION FIVE

> gear f

A machine tool is shown in Fig. 1. The motor shaft is connected to gear A and rotates

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the output shaft. What is the angular speed of the parallel shafts rotating together. The final gear F is

fixed to gear F? The number of teeth on each gear is given below.

See the figure for details

and

Number of teeth = r

Teeth 20 : 1

at SU : 2

— : 26 OS

r : (E+)

D =